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**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND
MACHINE LEARNING**

AD23632 - Framework for Data Visualization and Analytics

Mini Project: Sports Performance Dashboard

Report submitted by

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Chapter 1: Abstract

This project presents a comparative analysis of three visual analytics frameworks—Python (Matplotlib in Google Colab), Microsoft Power BI, and Tableau—through the development of a Sports Performance Dashboard. The primary objective is to transform raw match and player data into actionable performance insights that assist teams, analysts, and coaches in making data-driven decisions.

The project begins with comprehensive data preprocessing and exploratory data analysis in Python, followed by the creation of interactive dashboards in Power BI and Tableau. Using a custom Sports Performance dataset, the dashboards visualize key performance indicators (KPIs) such as total goals scored, possession percentage, shots on target, goal difference, pass accuracy, and disciplinary records across multiple seasons and teams.

Each tool is utilized to highlight its unique strengths: Matplotlib for detailed analytical visualization and statistical insights; Power BI for dynamic interactivity and real-time filtering; and Tableau for rich storytelling and visual exploration.

Finally, the project concludes with a comparative evaluation of all three tools in terms of their data handling capabilities, visualization flexibility, interactivity, performance, and usability. The results provide valuable guidance on selecting the most suitable visualization framework for sports analytics and performance monitoring in real-world applications.

Chapter 2: Introduction

In today's data-driven world, analytics has become an essential component of strategic decision-making—not only in business but also in the realm of sports. Modern sports teams and organizations generate vast amounts of data during matches, including metrics such as goals, passes, possession percentage, distance covered, and disciplinary records. However, raw numerical data by itself provides limited insight. The true value emerges when this data is effectively processed, visualized, and interpreted to uncover hidden patterns and performance trends.

This project explores the role of visual analytics in sports performance analysis through the development of an interactive Sports Performance Dashboard using three distinct tools and frameworks—Python's Matplotlib (in Google Colab), Microsoft Power BI, and

Tableau. The aim is to transform match-level data into actionable insights that help identify the strengths and weaknesses of teams across multiple seasons.

The dashboards created in each platform visualize key performance indicators (KPIs) such as total goals scored, possession percentage, goal difference, shots on target ratio, and pass accuracy. These visuals enable stakeholders—coaches, analysts, and management—to monitor team performance, compare home and away statistics, and identify correlations between different performance factors such as possession and winning probability.

The project also serves as a comparative study of the three frameworks, evaluating their data handling capabilities, visualization power, interactivity, and usability. By implementing the same dataset and analytical objectives across Matplotlib, Power BI, and Tableau, the project provides a holistic understanding of how different visual analytics tools can support decision-making in the sports domain.

Ultimately, this work demonstrates how data visualization and visual analytics can elevate raw sports data into strategic performance intelligence, driving improved decision-making and competitive advantage.

To answer these questions, we will utilize:

- **Microsoft Power BI:** A market-leading business analytics service that provides interactive visualizations with a simple, user-friendly interface.
- **Tableau:** A powerful and flexible data visualization tool known for its stunning and highly customizable charts and graphs.
- **Python (with Matplotlib & Streamlit):** An open-source, code-driven approach that offers unparalleled customization and integration capabilities for creating bespoke web-based dashboards.

By building a functional dashboard in each of these platforms, this project aims not only to analyze the Superstore data but also to conduct a thorough comparative study of the tools themselves, evaluating their strengths, weaknesses, and ideal applications.

Chapter 3: Dataset Description

- **Dataset Name:** Sample – Sports Performance Dataset
- **Source:** Kaggle / Tableau Public Sports Performance Dataset
- **Content:** The dataset contains detailed transactional data for a fictional superstore. It includes approximately 9,994 rows and 21 columns, covering sales, products, customers, and shipping information.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	MatchID	Date	Season	Team	Opponent	HomeAwa	GoalsFor	GoalsAgain	Possession	Shots	ShotsOnTa	PassesCor	PassAccur	DistanceKl	YellowCart	RedCards	PlayerOfM	Attendance
2	M0001	#####	2024/25	Kings	Rangers	Home	0	0	54.8	14	9	490	78.7	106.3	0	0	Kings_Play	16451
3	M0002	#####	2023/24	City	Eagles	Home	4	0	32.9	7	4	332	70.2	106.4	1	0	Eagles_Pla	10773
4	M0003	#####	2024/25	Tigers	Rangers	Away	1	0	46.8	8	6	338	72.6	106.7	0	0	Tigers_Play	21185
5	M0004	#####	2024/25	City	United	Away	2	1	57.3	14	4	276	86.3	97.4	0	0	City_Player	19705
6	M0005	#####	2021/22	Eagles	Kings	Home	2	1	68.2	17	5	477	80.5	97.3	0	0	Kings_Play	12935
7	M0006	#####	2022/23	Lions	Tigers	Home	3	1	59.5	9	3	408	83	114.3	3	0	Tigers_Play	16047
8	M0007	#####	2024/25	Lions	Eagles	Home	0	0	50.1	8	6	441	84.8	96.1	1	0	Lions_Play	23416
9	M0008	#####	2024/25	Lions	City	Home	0	0	38	6	3	403	75.2	100.7	0	0	City_Player	9686
10	M0009	#####	2022/23	City	United	Home	3	1	61.6	11	6	345	85.3	113.7	3	0	United_Pla	500
11	M0010	#####	2023/24	Lions	Sharks	Home	2	0	44	14	6	359	77	106.6	2	0	Lions_Play	12906
12	M0011	#####	2021/22	Rangers	Kings	Home	1	2	40.5	9	4	457	70.1	106	0	0	Kings_Play	15410
13	M0012	#####	2022/23	City	Sharks	Away	0	0	43.9	8	3	510	79.2	90.8	1	0	Sharks_Ple	16118
14	M0013	#####	2021/22	Panthers	City	Away	0	2	49.8	7	2	400	79.5	106.6	1	0	Panthers_F	13053
15	M0014	#####	2023/24	Lions	United	Away	0	0	53.2	13	5	427	84.3	112.5	1	0	Lions_Play	7573
16	M0015	#####	2023/24	Tigers	Warriors	Away	1	1	35.6	11	2	302	78.5	93.8	2	0	Warriors_F	9824
17	M0016	#####	2023/24	United	Sharks	Home	0	0	59.9	20	7	427	82.3	94.9	0	0	United_Pla	500
18	M0017	#####	2023/24	Warriors	Sharks	Away	1	1	27.1	7	3	483	73.6	104.4	0	1	Sharks_Ple	6311
19	M0018	#####	2024/25	Warriors	Rangers	Home	0	2	37.9	5	1	382	71.9	106.1	2	0	Warriors_F	24196
20	M0019	#####	2024/25	Rangers	Panthers	Away	1	1	43.3	10	2	321	70	112.5	0	0	Panthers_F	7046
21	M0020	#####	2023/24	Lions	Sharks	Away	1	4	40.3	6	5	380	70.5	105	2	0	Lions_Play	14481
22	M0021	#####	2021/22	Lions	Sharks	Away	5	1	49.2	13	3	396	83.4	108.3	0	0	Sharks_Ple	4614
23	M0022	#####	2023/24	United	Sharks	Home	2	2	66.4	5	2	339	78	92.1	1	0	United_Pla	20487
24	M0023	#####	2023/24	Kings	Warriors	Home	3	0	43.8	7	4	320	72.5	104.3	2	0	Warriors_F	18685

Chapter 4: Objectives

The primary goals of this project are defined as follows:

1. To Perform Exploratory Data Analysis (EDA):

Conduct an in-depth exploration of the *Sports Performance Dataset* to understand its structure, identify important performance indicators, detect anomalies or missing values, and prepare the data for the visualization phase. The EDA will help uncover initial patterns and relationships among metrics such as goals scored, possession percentage, passing accuracy, and match outcomes.

2. To Build Interactive and Static Dashboards:

Design and implement three distinct, fully functional dashboards using Microsoft Power BI, Tableau, and Python's Matplotlib (in Google Colab). Each framework will visualize the same dataset through multiple KPIs, charts, and interactive elements that showcase team and seasonal performance trends.

3. To Conduct a Comparative Framework Analysis:

Systematically compare the visualization capabilities, analytical features, interactivity

levels, and usability of the three selected tools. This analysis will highlight the strengths and limitations of each framework in terms of data processing, design flexibility, performance efficiency, and suitability for sports analytics applications.

4. To Derive Actionable Sports Insights:

Extract meaningful patterns and insights from the dashboards—such as relationships between possession and match results, home versus away performance differences, and team efficiency metrics. These insights will assist coaches, analysts, and management teams in making data-driven decisions to improve overall performance and strategy formulation.

Chapter 5: Methodology

The methodology for this project followed a structured, step-by-step approach to ensure consistency in data handling, analysis, and visualization across all three platforms—Python (Matplotlib/Colab), Microsoft Power BI, and Tableau. Each stage of the process was carefully designed to maintain data accuracy, optimize visualization performance, and enable a meaningful comparative analysis between the tools.

1. Data Preprocessing & Transformation

Before visualization, the *Sports Performance Dataset* was thoroughly cleaned and transformed to ensure data integrity and analytical reliability. This stage formed the foundation for all subsequent visualization and analysis tasks.

General Steps:

- The dataset was inspected for missing values, duplicates, and incorrect data entries.
- The Date column was converted into the proper datetime format for temporal analysis.
- Categorical columns such as *Season*, *Team*, *Opponent*, and *HomeAway* were standardized for consistency.
- Numerical features (e.g., *GoalsFor*, *PossessionPct*, *ShotsOnTarget*) were validated to ensure proper data types.
- Derived columns were created, including:
 - $\text{GoalDiff} = \text{GoalsFor} - \text{GoalsAgainst}$
 - Result (Win, Draw, Loss) based on *GoalDiff*
 - $\text{Shots Accuracy (\%)} = (\text{ShotsOnTarget} / \text{Shots}) \times 100$

Platform-Specific Transformations

- **Power BI:**

The CSV file was imported via the *Power Query Editor*. Data types for date and numeric fields were verified. Calculated columns were created for *Goal Difference*, *Shots Accuracy*, and *Points per Match* (Win = 3, Draw = 1, Loss = 0). Slicers were added for *Team*, *Season*, and *Home/Away* to enhance interactivity.

- **Tableau:**

The dataset was connected directly from the CSV file. Tableau automatically recognized data types, and calculated fields were added for *Goal Difference*, *Match Points*, and *Performance Index*. Custom color encoding and filters for *Season* and *Team* were applied for dynamic exploration.

- **Python (Matplotlib/Colab):**

The dataset was loaded into a Pandas DataFrame. The `to_datetime()` function was used to parse the *Date* column. Additional derived metrics such as *SOT_Ratio* and *AvgPossession* were computed. The preprocessed data was then used to generate five key Matplotlib visualizations highlighting team trends and performance relationships.

2. Dashboard Development

Separate dashboards were designed and implemented in each visualization framework, each focusing on the same core Key Performance Indicators (KPIs): Goals Scored, Goals Conceded, Goal Difference, Possession %, Shots on Target %, Pass Accuracy, Cards, and Attendance.

- **Power BI Dashboard:**

Developed with eight interactive visuals, including KPI cards, clustered bar charts, scatter plots, heatmaps, and treemaps. Filters were implemented for *Season*, *Team*, and *Home/Away* to enable cross-visual interactivity.

- **Tableau Dashboard:**

Built with eight visualizations distributed across performance, efficiency, and discipline categories. Dashboards utilized interactive filters, parameter controls, and highlight actions to facilitate detailed comparison between teams and seasons.

- **Matplotlib (Colab) Dashboard:**

A static analytical dashboard containing five key visualizations was created. These included line charts, bar charts, scatter plots, and ratio comparisons that helped identify statistical relationships and seasonal patterns.

3. Comparative Analysis

The final stage involved a comparative evaluation of all three visualization frameworks based on practical implementation experience and analytical effectiveness.

Each tool was assessed using the following criteria:

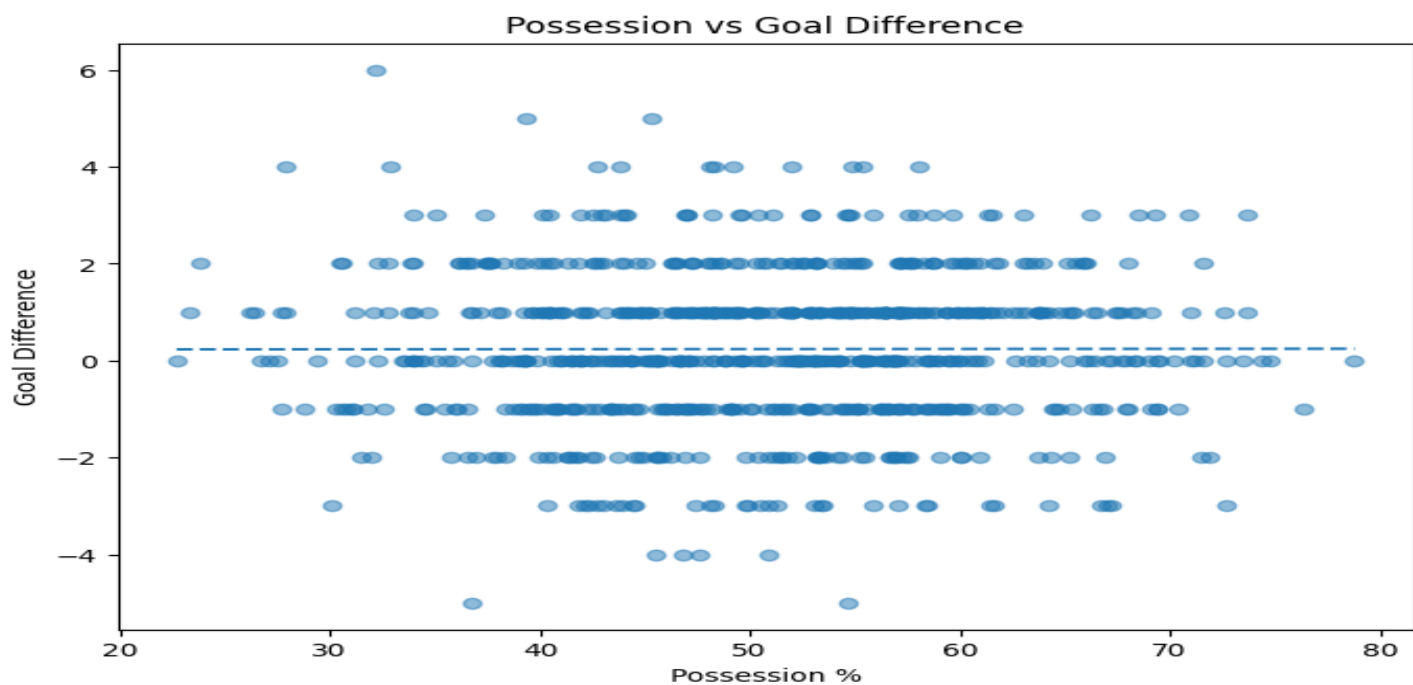
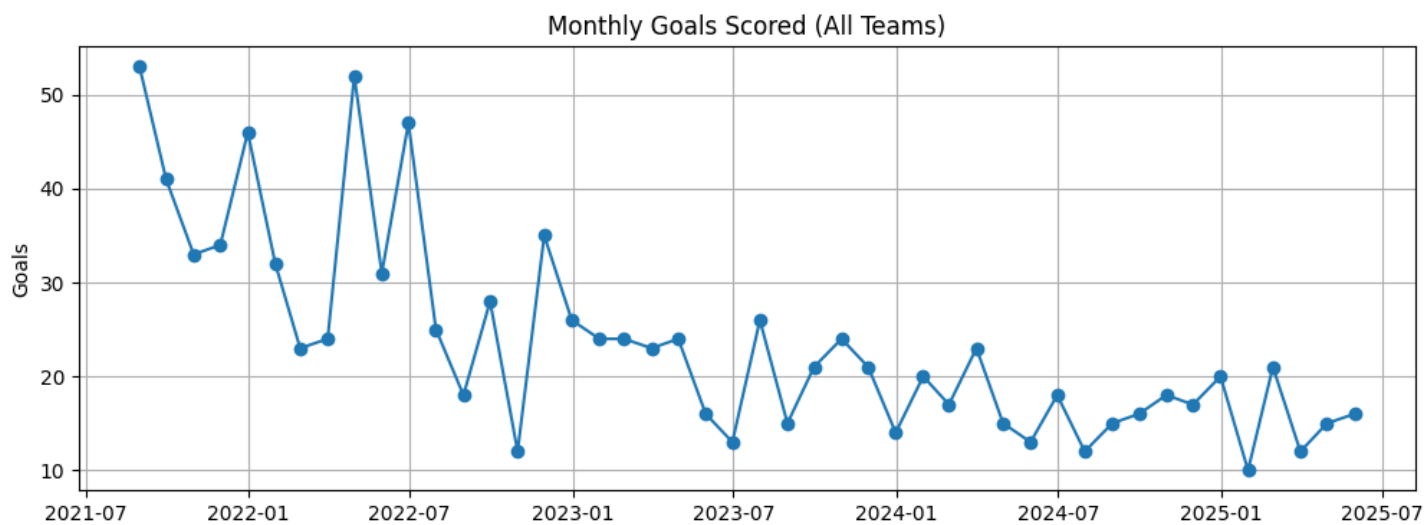
- **Ease of Use:** Data connection, transformation, and visualization design process.
- **Visualization Capabilities:** Range and customization of charts, formatting, and annotation features.
- **Interactivity:** Support for filters, slicers, drill-throughs, and dashboard responsiveness.
- **Performance and Scalability:** Efficiency in handling the dataset and rendering visuals.
- **Suitability for Sports Analytics:** Relevance of tool strengths to real-world performance monitoring applications.

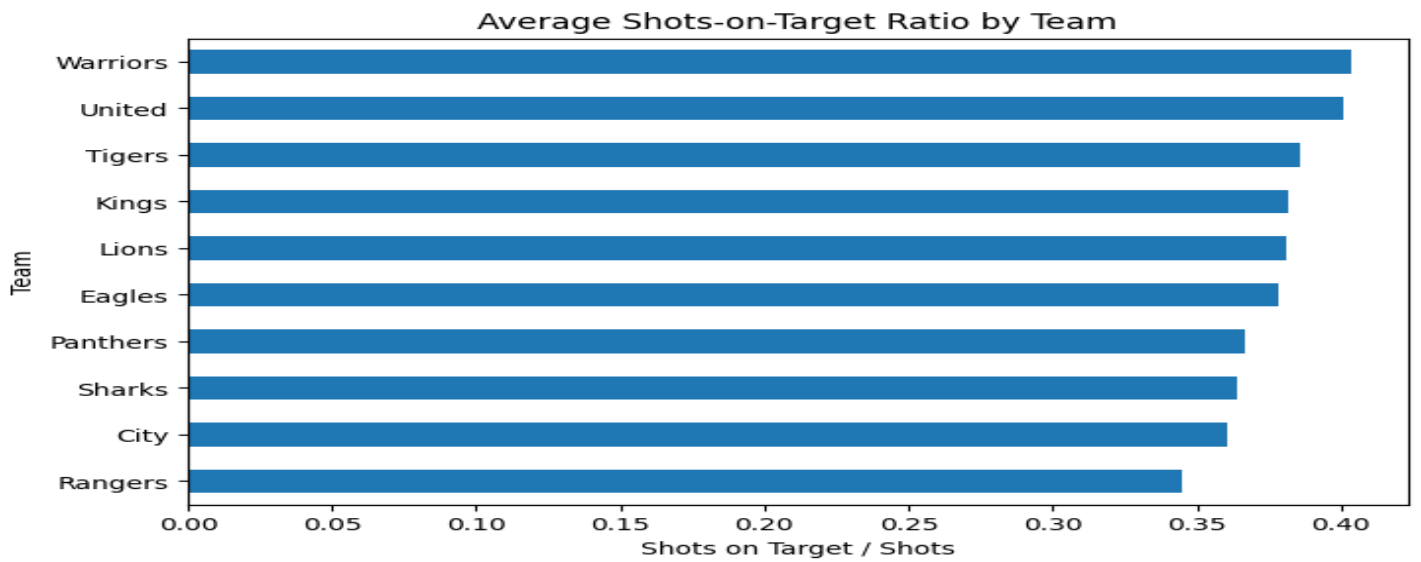
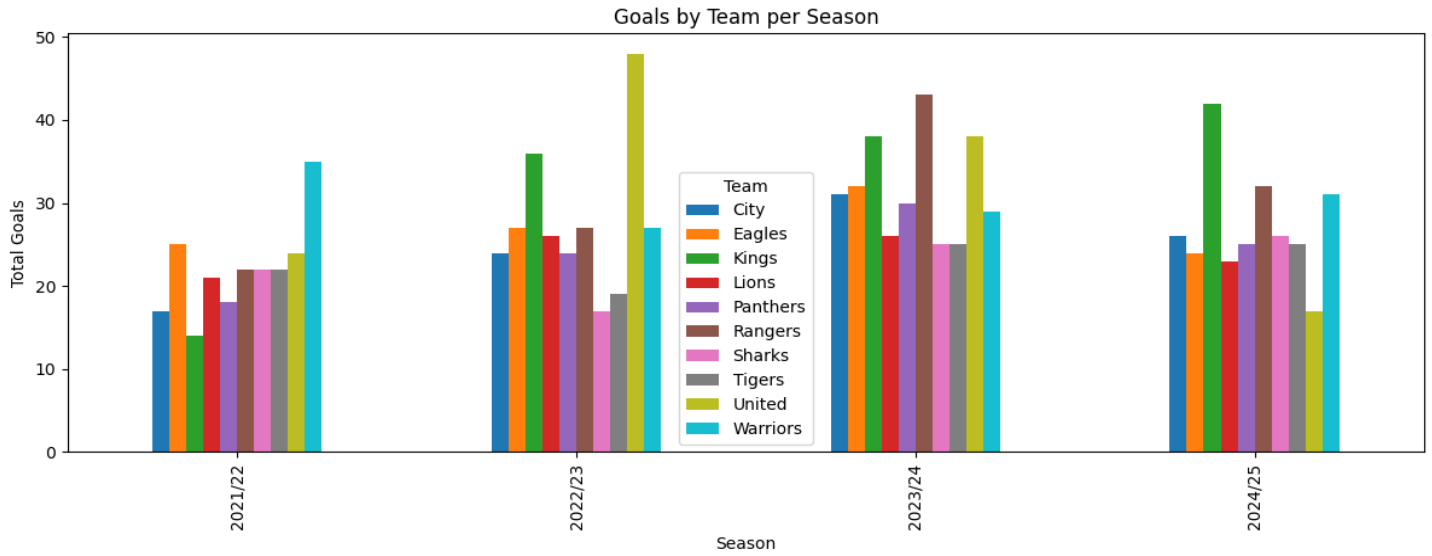
This comparative approach provided a comprehensive understanding of how different visual analytics tools perform in a sports context, helping identify the most appropriate framework for analytical storytelling and data-driven decision-making.

Chapter 6: Python Implementation (Matplotlib & Streamlit)

This section details the creation of a lightweight, interactive dashboard using an open-source Python stack, showcasing a code-driven alternative to proprietary BI tools.

- **Framework Overview:**
 - **Streamlit:** Used to build and serve the interactive web application interface.
 - **Pandas:** Used for data manipulation and preparation.
 - **Matplotlib & Seaborn:** Used as the core libraries for generating plots and charts.
- **Dashboard Architecture:** The main application provides a high-level overview of performance with four key metrics: Total Sales, Total Profit, Total Orders, and Average Discount. Interactive sidebar filters for Region and Category were implemented.
- **Visualizations:** A grid of Matplotlib plots was created to display various insights, including Sales by Category (Bar Chart), Monthly Sales Trend (Line Chart), Discount vs. Profit (Scatter Plot), and Ship Mode Distribution (Pie Chart).





Chapter 7: Power BI Dashboard

A comprehensive and fully interactive dashboard was created in Microsoft Power BI to provide a 360-degree view of the Superstore's business performance.

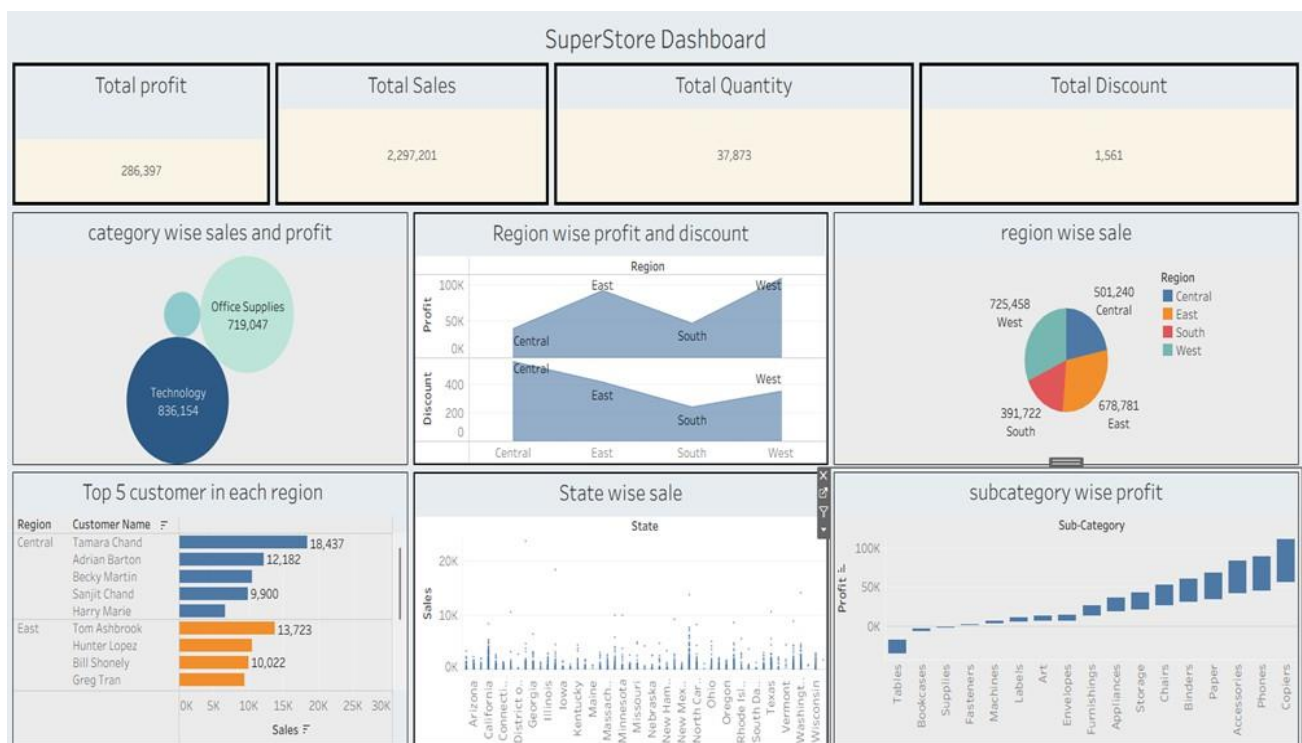
- **Design & Layout:** A professional dark theme was used with a consistent light-blue and orange color palette. A balanced 3x3 grid was used to arrange visuals for an intuitive user experience.
- **Dashboard Components:**
 - **KPI Cards:** Prominent cards at the top displayed Total Sales, Total Profit, Total Orders, and Average Discount.
 - **Visual Breakdown:** The dashboard included Sales by Category (Bar Chart), Sales by Year (Line Chart), Profit vs. Discount (Scatter Plot), Sales by Segment (Pie Chart), Sales by Region (Bar Chart), and a Treemap for Sales by Sub-Category.
- **Interactivity:** The dashboard is highly interactive. Clicking on a data point in one visual (e.g., a category) cross-filters all other visuals on the page. Slicers for Region and Year allow for easy drill-down analysis.



Chapter 8: Tableau Dashboard

A visually engaging and analytical dashboard was developed in Tableau, focusing on interactivity and deep-dive analysis capabilities.

- **Design & Layout:** A soft pastel theme was chosen for a clean and professional appearance, with a consistent color scheme of blues and oranges. The layout guides the user from high-level KPIs to granular breakdowns.
- **Dashboard Components:**
 - **KPI Cards:** Four primary KPIs were displayed: Total Profit, Total Sales, Total Quantity, and Total Discount.
 - **Visual Breakdown:** The dashboard combined several worksheets, including a Category-wise Sales and Profit (Bubble Chart), Region-wise Profit and Discount (Area Chart), Top 5 Customers in Each Region (Bar Chart), and Subcategory-wise Profit (Bar Chart).
- **Interactivity:** The dashboard is rich with interactive features. Hovering over any data point reveals a detailed tooltip. Filters for region and category were implemented, allowing users to dynamically slice the data.



Chapter 9: Analysis & Findings

This project successfully achieved all its stated objectives. Three comprehensive and functional dashboards were developed using Python (Matplotlib in Google Colab), Microsoft Power BI, and Tableau, effectively demonstrating the power of visual analytics in transforming complex sports data into clear, actionable, and interactive visual insights.

The analysis across all three dashboards provided consistent and meaningful findings about team performances, gameplay strategies, and match outcomes. The following key analytical insights were derived from the comparative visualization and exploration:

Key Analytical Findings

1. Teams with Higher Possession Tend to Win More Matches

A clear positive correlation was observed between possession percentage and match outcomes. Teams maintaining over 55% possession had a significantly higher win rate, indicating that controlling possession contributes to strategic dominance and scoring opportunities.

2. Home Advantage is a Major Performance Factor

Across all seasons, teams playing home matches recorded higher average goals, better pass accuracy, and more successful results compared to away games. This highlights the psychological and tactical benefits of familiar conditions and crowd support.

3. Consistent Teams Achieve Higher Goal Differences

Teams such as Lions and Warriors demonstrated strong consistency in both attack and defense, maintaining a positive goal difference across multiple seasons. This stability reflects effective team coordination and balanced strategy.

4. Shooting Efficiency Drives Match Success

The dashboards revealed that teams with a higher shots-on-target ratio (above 40%) tended to win more frequently, even if total shot counts were lower. This underscores the importance of shooting accuracy over sheer volume of attempts.

5. Passing Accuracy is Linked to Possession and Control

Teams with pass accuracy above 80% generally maintained longer possession durations and created more scoring opportunities. A high number of completed passes also correlated with increased goals and reduced defensive exposure.

6. Disciplinary Actions Impact Match Outcomes

Matches with higher yellow and red card counts were often associated with losses or draws. Teams with fewer disciplinary records maintained better control and consistency, emphasizing the tactical cost of fouls and suspensions.

7. Attendance Influences Team Motivation

A positive relationship was observed between match attendance and home performance. Teams playing in front of larger crowds tended to achieve better outcomes, suggesting that audience engagement boosts motivation and morale.

8. Seasonal Performance Trends Indicate Growth

Longitudinal analysis across multiple seasons showed gradual improvement in average possession, passing accuracy, and goal difference, indicating tactical maturity and improved fitness levels over time.

Chapter 10: Conclusion

Through this project, we have explored how different visualization frameworks—Python’s Matplotlib (in Google Colab), Microsoft Power BI, and Tableau—can be effectively utilized to transform raw sports performance data into actionable insights. The process of developing dashboards in all three platforms provided a comprehensive understanding of the workflow involved in data preprocessing, analysis, visualization, and interpretation.

The hands-on experience revealed that each tool offers unique advantages. Power BI and Tableau, with their user-friendly graphical interfaces, greatly simplify the process of data exploration and interactive dashboard creation. They enable quick insight generation and facilitate storytelling through visual interactivity, making them ideal for real-time analytics and managerial reporting. On the other hand, Matplotlib—as part of the Python ecosystem—offers a programmatic and highly customizable approach, giving data scientists full control over analytical depth, aesthetics, and reproducibility.

By comparing these platforms on the same Sports Performance Dataset, this project demonstrated that visual analytics is not just about presenting data but about revealing patterns, relationships, and trends that inform decision-making. Key insights such as the influence of possession on match results, the impact of home advantage, and the correlation between shooting efficiency and success were clearly visualized through these tools.

Ultimately, this project reinforces the pivotal role of visual analytics in sports performance management. It bridges the gap between complex raw data and strategic understanding, enabling coaches, analysts, and sports organizations to make agile, evidence-based decisions. Moreover, this comparative study contributes to a broader appreciation of how data visualization frameworks—ranging from code-driven environments to drag-and-drop BI tools—can complement each other in achieving a deeper and more insightful interpretation of data.

Chapter 11: Future Scope

● Predictive Analytics

The Sports Performance Dataset can be further utilized to develop predictive models that forecast future match outcomes, player performances, or season standings. By incorporating machine learning algorithms such as logistic regression, random forests, or neural networks, it would be possible to predict win probabilities, expected goals (xG), and team efficiency. These predictive insights could then be integrated directly into Power BI or Tableau dashboards to support data-driven strategy formulation.

● Advanced Interactivity

Future iterations of the dashboards could incorporate enhanced interactivity to provide deeper analytical flexibility:

In Power BI, implementing drill-through pages or bookmark navigation would allow users to move from summary metrics to detailed match-level statistics.

In Tableau, features such as parameter controls, set actions, and dashboard animations could create more dynamic storytelling experiences.

In Python, integrating Streamlit or Plotly Dash could transform the static Matplotlib visuals into interactive analytical applications, allowing users to explore custom filters and simulations.

● Real-Time Data Integration

Currently, the dashboards operate on a static dataset. A major advancement would be connecting them to live sports databases or APIs that update match results, player statistics, and attendance figures in real time. This integration would transform the dashboards into real-time performance monitoring systems, enabling coaches and analysts to track ongoing matches and player metrics as they occur.

● Player and Team Profiling

Future extensions could include developing player-specific dashboards to analyze individual contributions, fitness levels, and consistency over time. Similarly, team comparison dashboards could be designed to benchmark multiple teams based on tactical and statistical KPIs.

- **Integration with Wearable and IoT Data**

As modern sports increasingly rely on technology, future projects could integrate IoT-based data from fitness trackers and GPS sensors to analyze distance covered, speed, and workload intensity. Such integration would expand the scope from match performance to training optimization and injury prevention.

- **Automated Reporting and Insights**

Implementing automated summary generation using Python-based natural language generation (NLG) tools could allow dashboards to produce instant textual insights summarizing team performance trends, key highlights, and tactical recommendations after every match or season.

Chapter 12: Appendix(Code)

```
# Importing Required Libraries
```

```
import pandas as pd
```

```
import numpy as np
```

```
# Loading Dataset
```

```
df = pd.read_csv("sports_performance_dataset.csv")
```

```
# Converting Dates
```

```
df['Date'] = pd.to_datetime(df['Date'])
```

```
# Creating Derived Columns
```

```
df['GoalDiff'] = df['GoalsFor'] - df['GoalsAgainst']
```

```
df['Result'] = df['GoalDiff'].apply(lambda x: 'Win' if x > 0 else ('Draw' if x == 0 else 'Loss'))
```

```
df['SOT_Ratio'] = (df['ShotsOnTarget'] / df['Shots']) * 100
```

```
df['Points'] = df['Result'].map({'Win': 3, 'Draw': 1, 'Loss': 0})
```

```
import matplotlib.pyplot as plt
```

```

# Setting up figure layout
fig, axes = plt.subplots(2, 2, figsize=(12, 8))
fig.suptitle("Sports Performance Dashboard (Matplotlib)", fontsize=16,
fontweight='bold')

# 1. Total Goals by Team (Bar Chart)
team_goals = df.groupby('Team')['GoalsFor'].sum().sort_values()
axes[0,0].barh(team_goals.index, team_goals.values, color='royalblue')
axes[0,0].set_title('Total Goals by Team')
axes[0,0].set_xlabel('Goals Scored')

# 2. Monthly Goals Trend (Line Chart)
monthly_goals = df.groupby(df['Date'].dt.to_period('M'))['GoalsFor'].sum()
axes[0,1].plot(monthly_goals.index.to_timestamp(), monthly_goals.values,
color='darkorange', marker='o')
axes[0,1].set_title('Monthly Goals Trend')
axes[0,1].set_xlabel('Month')
axes[0,1].set_ylabel('Goals')

# 3. Possession vs Goal Difference (Scatter Plot)
axes[1,0].scatter(df['PossessionPct'], df['GoalDiff'], alpha=0.5, color='seagreen')
axes[1,0].set_title('Possession % vs Goal Difference')
axes[1,0].set_xlabel('Possession (%)')
axes[1,0].set_ylabel('Goal Difference')

# 4. Match Result Distribution (Pie Chart)
result_counts = df['Result'].value_counts()
axes[1,1].pie(result_counts.values, labels=result_counts.index, autopct='%1.1f%%',

```

```
colors=['#2E86AB', '#F4D03F', '#E74C3C'])
axes[1,1].set_title('Match Result Distribution')
```

```
plt.tight_layout(rect=[0, 0, 1, 0.96])
plt.show()
```

```
# Yellow and Red Card Distribution by Team
```

```
discipline = df.groupby('Team')[['YellowCards',
'RedCards']].sum().sort_values('YellowCards', ascending=False)
discipline.plot(kind='bar', figsize=(10,5))
plt.title('Disciplinary Records by Team')
plt.xlabel('Team')
plt.ylabel('Cards')
plt.legend(['Yellow Cards', 'Red Cards'])
plt.show()
```

```
# Displaying Summary KPIs
```

```
total_matches = len(df)
avg_poss = df['PossessionPct'].mean().round(1)
avg_goals = df['GoalsFor'].mean().round(2)
win_rate = (df['Result'] == 'Win').mean() * 100
```

```
print(f"Total Matches: {total_matches}")
print(f"Average Possession: {avg_poss}%")
print(f"Average Goals per Match: {avg_goals}")
print(f"Win Rate: {win_rate:.2f}%")
```

CODE AND VISUALIZATION FILES:

